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Kelas / Absen: TI_1G / 20

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Tugas Jobsheet 14

BinaryTreeMain20.java:

```
bst.addRekursif(new Mahasiswa20("244160121", "Ali", "A", 3.57));  
bst.addRekursif(new Mahasiswa20("244160221", "Badar", "B", 3.85));  
bst.addRekursif(new Mahasiswa20("244160185", "Candra", "C", 3.21));  
bst.addRekursif(new Mahasiswa20("244160220", "Dewi", "B", 3.54));  
  
bst.cariMaxIPK();  
bst.cariMinIPK();  
  
System.out.println("\nnipk di atas");  
  
bst.tampilMahasiswaIPKDiAtas(bst.root, 3.50);
```

Tugas 1:

```
bst.addRekursif(new Mahasiswa20("244160121", "Ali", "A", 3.57));  
bst.addRekursif(new Mahasiswa20("244160221", "Badar", "B", 3.85));  
bst.addRekursif(new Mahasiswa20("244160185", "Candra", "C", 3.21));  
bst.addRekursif(new Mahasiswa20("244160220", "Dewi", "B", 3.54));
```

Hasil:

```
Daftar semua Mahasiswa (In Order Traverse):  
NIM   : 244160185  
Nama  : Candra  
Kelas: C  
IPK   : 3.21  
=====
```

NIM	Nama	Kelas	IPK
244160185	Candra	C	3.21
244160220	Dewi	B	3.54
244160121	Ali	A	3.57
244160221	Badar	B	3.85

```
=====
```

Tugas 2:

```
void cariMinIPK(){  
    if (isEmpty()) {  
        System.out.println("Binary Tree masih kosong");  
        return;  
    }  
    Node20 current = root;  
    while (current.left != null) {  
        current = current.left;  
    }  
}
```

```

        System.out.println("Min IPK: ");
        current.mahasiswa.tampilInformasi();
    }

    void cariMaxIPK(){
        if (isEmpty()) {
            System.out.println("Binary Tree masih kosong");
            return;
        }
        Node20 current = root;
        while (current.right != null) {
            current = current.right;
        }

        System.out.println("Max IPK: ");
        current.mahasiswa.tampilInformasi();
    }

```

Hasil:

```

Max IPK:
NIM   : 244160221
Nama  : Badar
Kelas: B
IPK   : 3.85
=====
Min IPK:
NIM   : 244160185
Nama  : Candra
Kelas: C
IPK   : 3.21
=====

```

Tugas 3:

```
void tampilMahasiswaIPKDiAtas(Node20 node, double input){  
    if (node != null) {  
        tampilMahasiswaIPKDiAtas(node.left, input);  
        if (node.mahasiswa.ipk > input) {  
            node.mahasiswa.tampilInformasi();  
        }  
        tampilMahasiswaIPKDiAtas(node.right, input);  
    }  
}
```

Hasil:

```

ipk di atas
NIM   : 244160220
Nama  : Dewi
Kelas: B
IPK   : 3.54
=====
NIM   : 244160131
Nama  : Devi
Kelas: A
IPK   : 3.72
=====
NIM   : 244160221
Nama  : Badar
Kelas: B
IPK   : 3.85
=====

```

Tugas 4:

BinaryTreeArrayMain20.java:

```

bta.add(new Mahasiswa20("244160110", "Akmal", "G", 3.67));
System.out.println("\nPreOrder Traversal Mahasiswa: ");
bta.traversePreOrder(0);

```

BinaryTreeArray20.java:

```

void add(Mahasiswa20 mahasiswa20){
    if (id < dataMahasiswa.length) {
        id++;
        dataMahasiswa[id] = mahasiswa20;
    }
}

```

```

void traversePreOrder(int idxStart){
    if (idxStart <= idxLast) {

```

```
    if (dataMahasiswa[idxStart] != null) {  
        dataMahasiswa[idxStart].tampilInformasi();  
        traverseInOrder(2*idxStart+1);  
        traverseInOrder(2*idxStart+2);  
    }  
}  
}
```

Hasil:

PreOrder Traversal Mahasiswa:

NIM : 244160121

Nama : Ali

Kelas: A

IPK : 3.57

=====

NIM : 244160220

Nama : Dewi

Kelas: B

IPK : 3.35

=====

NIM : 244160110

Nama : Akmal

Kelas: G

IPK : 3.67

=====

NIM : 244160131

Nama : Devi

Kelas: A

IPK : 3.48

=====

NIM : 244160205

Nama : Ehsan

Kelas: D

IPK : 3.61

=====

NIM : 244160185

Nama : Badar

Kelas: C

IPK : 3.75

=====

NIM : 244160170

Nama : Fizi

Kelas: B

IPK : 3.86

=====